

半导体中载流子的统计分布

如何求n0?----

$$n_0 = \frac{1}{\text{体积}} \int_{E_c}^{E'_c} \text{状态密度} \times \text{概率} \, dE$$

状态密度: $g_c(E) = \frac{V}{2\pi^2} \frac{(2m_n^*)^{\frac{3}{2}}}{\hbar^3} (E - E_c)^{\frac{1}{2}}$

简并半导体

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$$

如何判断是否简并

n型半导体

- $E_c - E_F > 2k_0T$ 非简并
- $0 < E_c - E_F \leq 2k_0T$ 弱简并
- $E_c - E_F \leq 0$ 简并

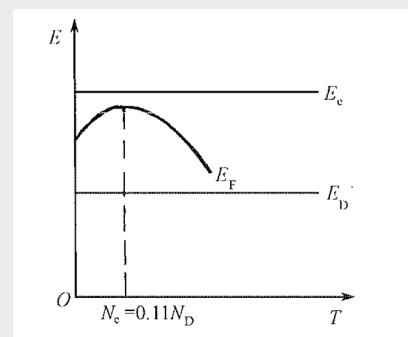
p型半导体

- $E_F - E_v > 2k_0T$ 非简并
- $0 < E_F - E_v \leq 2k_0T$ 弱简并
- $E_F - E_v \leq 0$ 简并

如何防止简并

$$N_0 \leq \frac{2}{0.11} \left(\frac{m_n^* \Delta E_D}{\pi \hbar^2} \right)^{\frac{3}{2}}$$

思路: 当费米能级EF最大时, 仍不满足简并的条件, 则全温度区间不满足简并条件



由 $n_0 = \frac{1}{V} \int_{E_c}^{E'_c} f_B(E) g_c(E) dE$

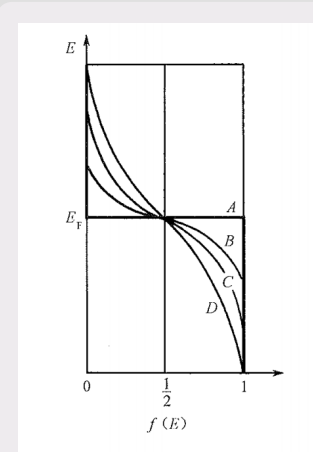
$$n_0 = N_c e^{-\frac{E_c - E_F}{kT}}$$
$$p_0 = N_v e^{-\frac{E_F - E_v}{kT}}$$

$$n_i^2 = n_0 p_0 = N_c N_v \exp\left(-\frac{E_g}{k_0T}\right)$$

给定半导体材料, ni只与温度有关

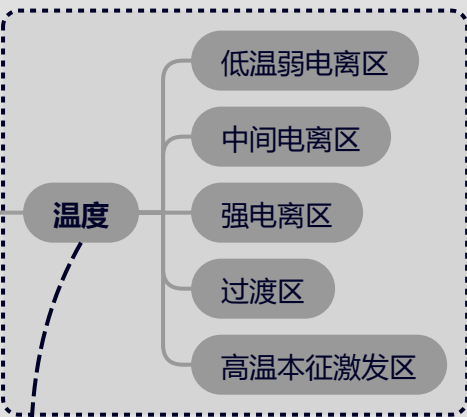
非简并半导体

$$f_B(E) = e^{-\frac{E - E_F}{k_0T}}$$



费米能级: 电子填充能量状态水平的分界线

均以n型半导体为例



无掺杂 (本征半导体)

全温度区间

$$n_0 = p_0$$

$$\begin{cases} n_0 = N_c e^{-\frac{E_c - E_F}{k_0T}} \\ n_0 = p_0 \end{cases}$$

费米能级 Ef

$$E_F = \frac{E_c + E_v}{2} + \frac{k_0T}{2} \ln \frac{N_v}{N_c}$$

导带电子浓度 n0

$$n_0 = n_i = (N_c N_v)^{\frac{1}{2}} e^{-\frac{E_g}{2k_0T}}$$

低温弱电离区

$$n_0 = n_D^+, n_D^+ \ll N_D$$

$$\begin{cases} n_0 = N_c e^{-\frac{E_c - E_F}{k_0T}} \\ n_0 = n_D^+, n_D^+ \ll N_D \end{cases}$$

费米能级 Ef

$$E_F = \frac{E_c + E_D}{2} + \frac{k_0T}{2} \ln \left(\frac{N_D}{2N_c} \right)$$

导带电子浓度 n0

$$n_0 = \left(\frac{N_D N_c}{2} \right)^{\frac{1}{2}} e^{-\frac{\Delta E_D}{2k_0T}}$$

中间电离区

$$n_0 = n_D^+$$

$$\begin{cases} n_0 = N_c e^{-\frac{E_c - E_F}{k_0T}} \\ n_0 = n_D^+ \end{cases}$$

费米能级 Ef

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导带电子浓度 n0

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强电离区

$$n_0 = N_D$$

$$\begin{cases} n_0 = N_c e^{-\frac{E_c - E_F}{k_0T}} \\ n_0 = N_D \end{cases}$$

费米能级 Ef

$$E_F = E_c + k_0T \ln \frac{N_D}{N_c}$$

导带电子浓度 n0

$$n_0 = N_D$$

过渡区

$$n_0 = p_0 + N_D$$

$$\begin{cases} n_0 = N_c e^{-\frac{E_c - E_F}{k_0T}} \\ n_0 = p_0 + N_D \end{cases}$$

费米能级 Ef

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导带电子浓度 n0

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高温本征激发区

$$n_0 = p_0 = n_i$$

$$\begin{cases} n_0 = N_c e^{-\frac{E_c - E_F}{k_0T}} \\ n_0 = p_0 \end{cases}$$

费米能级 Ef

$$E_F = \frac{E_c + E_v}{2} + \frac{k_0T}{2} \ln \frac{N_v}{N_c}$$

导带电子浓度 n0

$$n_0 = n_i = (N_c N_v)^{\frac{1}{2}} e^{-\frac{E_g}{2k_0T}}$$

多掺杂

$$n_0 = (N_D - N_A - n_D) + p_0$$

低温弱电离

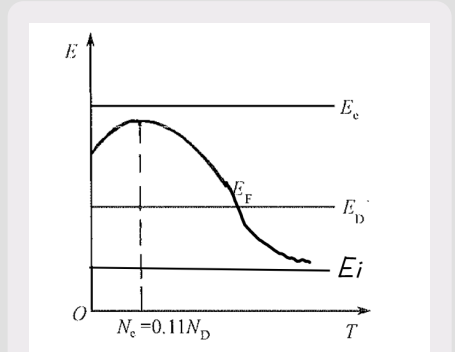
$$E_F = E_D + k_0T \ln \left(\frac{N_D - N_A}{2N_A} \right)$$

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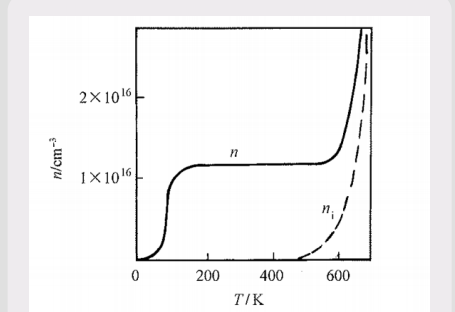
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形式与单一掺杂相似



EF 与温度 T



n0 与温度 T